Wireshark Lab: DNS v8.1
Questions and Answers

1. Run nslookup to obtain the IP address of the web server for the Indian Institute of Technology in Bombay, India: www.iitb.ac.in. What is the IP address of www.iitb.ac.in?

Answer: 103.21.124.10.

2. What is the IP address of the DNS server that provided the answer to your nslookup command in question 1 above?

Answer: 75.75.75.75.

3. Did the answer to your nslookup command in question 1 above come from an authoritative or non-authoritative server?

Answer: The answer came from a non-authoritative name server.

4. Use the nslookup command to determine the name of the authoritative name server for the iitb.ac.in domain. What is that name? (If there are more than one authoritative servers, what is the name of the first authoritative server returned by nslookup)? If you had to find the IP address of that authoritative name server, how would you do so?

Answer: dns1.iitb.ac.in. To find its IP address, run nslookup dns1.iitb.ac.in.

5. Locate the first DNS query message resolving the name gaia.cs.umass.edu. What is the packet number in the trace for the DNS query message? Is this query message sent over UDP or TCP?

Answer: Packet 15, sent over UDP.

6. Now locate the corresponding DNS response to the initial DNS query. What is the packet number in the trace for the DNS response message? Is this response message received via UDP or TCP?

Answer: Packet 17, received over UDP.

7. What is the destination port for the DNS query message? What is the source port of the DNS response message?

Answer: The query destination port is 53, and the response source port is 53.

8. To what IP address is the DNS query message sent?

Answer: 75.75.75.75.

9. Examine the DNS query message. How many "questions" does this DNS message contain? How many "answers" answers does it contain?

Answer: One question and zero answers.

10. Examine the DNS response message to the initial query message. How many "questions" does this DNS response message contain? How many "answers" answers does it contain?

Answer: One question and one answer.

11. The web page for the base file http://gaia.cs.umass.edu/kurose_ross/index.php references the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_9E_1.jpg, which, like the base webpage, is on gaia.cs.umass.edu. What is the packet number in the trace for the initial HTTP GET request for the base file http://gaia.cs.umass.edu/kurose_ross/? What is the packet number in the trace of the DNS query made to resolve gaia.cs.umass.edu so that this initial HTTP request can be sent to the gaia.cs.umass.edu IP address? What is the packet number in the trace of the received DNS response? What is the packet number in the trace for the HTTP GET request for the image object http://gaia.cs.umass.edu/kurose_ross/header_graphic_book_9E1.jpg? What is the packet number in the DNS query made to resolve gaia.cs.umass.edu so that this second HTTP request can be sent to the gaia.cs.umass.edu IP address? Discuss how DNS caching affects the answer to this last question.

Answer: Base-page GET: packet 22. DNS query: packet 15. DNS response: packet 17. The image GET is packet 205 for header_graphic_book_8E_2.jpg in the supplied trace; there is no GET for the 9E image filenames printed in the question. No new DNS query is sent for the image because the host reuses the cached address for gaia.cs.umass.edu.

12. What is the destination port for the DNS query message? What is the source port of the DNS response message?

Answer: The query destination port is 53, and the response source port is 53.

13. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

Answer: 75.75.75.75. Yes; it is the default local DNS server of the host shown in the question.

14. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"?

Answer: Type A, which requests an IPv4 address. The query contains no answers.

15. Examine the DNS response message to the query message. How many "questions" does this DNS response message contain? How many "answers"?

Answer: One question and one answer. The answer gives the IPv4 address 128.119.240.84 for www.cs.umass.edu.

16. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server?

Answer: 75.75.75.75. Yes; it is the default local DNS server of the host in the supplied trace.

17. Examine the DNS query message. How many questions does the query have? Does the query message contain any "answers"?

Answer: One question. The query contains no answers.

18. Examine the DNS response message (in particular the DNS response message that has type "NS"). How many answers does the response have? What information is contained in the answers? How many additional resource records are returned? What additional information is included in these additional resource records (if additional information is returned)?

Answer: Three NS answers give the authoritative name-server names ns1.umass.edu, ns2.umass.edu and ns3.umass.edu. Three additional A records give their IPv4 addresses:
ns1.umass.edu: 128.119.10.27
ns2.umass.edu: 128.119.10.28
ns3.umass.edu: 128.103.38.68

报告成绩查询

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提取码：B7J8